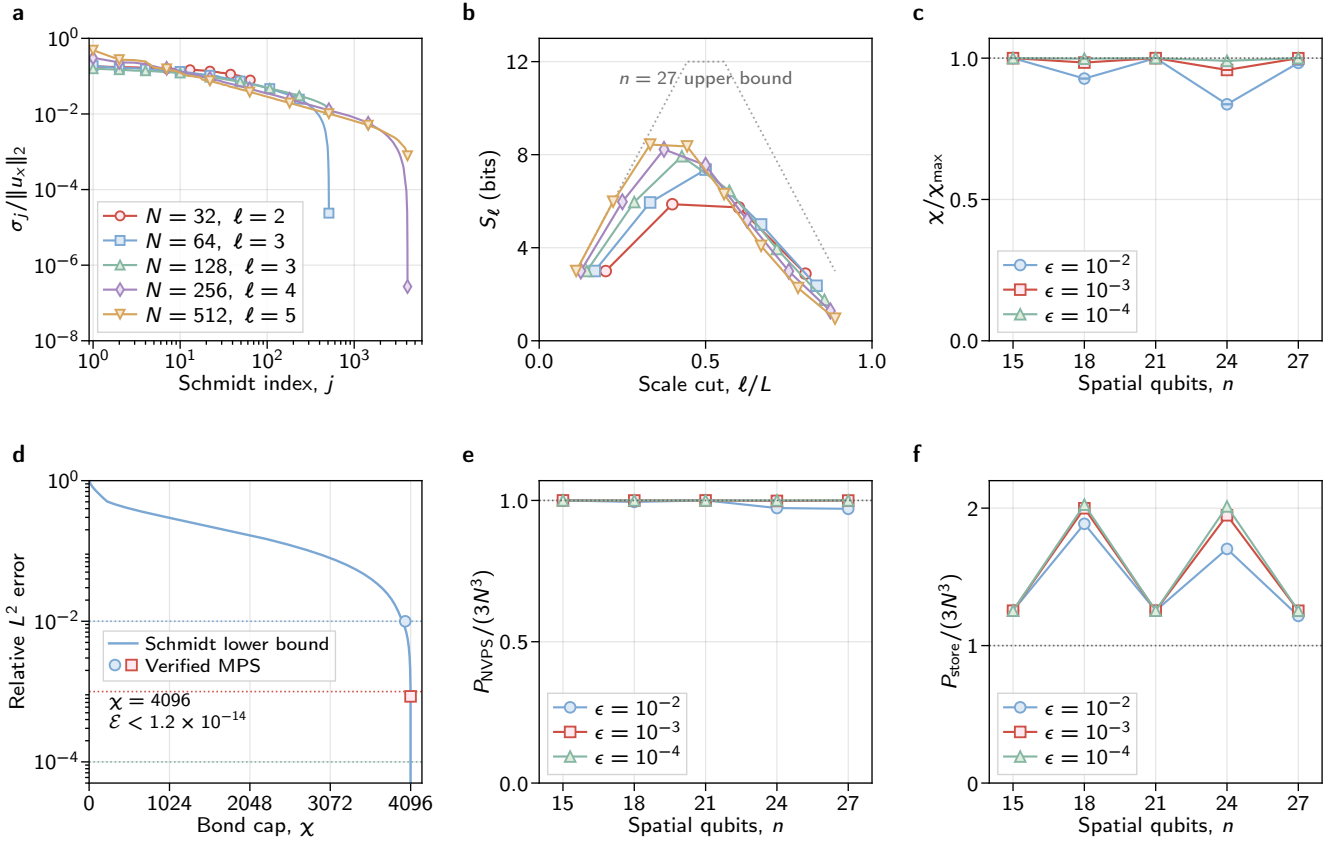

a
$\sigma_j/\|u_x\|_2$
Schmidt index, j
$N = 32,\ \ell = 2$
$N = 64,\ \ell = 3$
$N = 128,\ \ell = 3$
$N = 256,\ \ell = 4$
$N = 512,\ \ell = 5$
b
S_ℓ (bits)
Scale cut, ℓ/L
$n = 27$: upper bound
c
$\chi/\chi_{\max}$
Spatial qubits, n
$\epsilon = 10^{-2}$
$\epsilon = 10^{-3}$
$\epsilon = 10^{-4}$
d
Relative L^2 error
Bond cap, χ
Schmidt lower bound
Verified MPS
$\chi = 4096$
$\mathcal{E} < 1.2 \times 10^{-14}$
e
$P_{\text{NVPS}}/(3N^3)$
Spatial qubits, n
f
$P_{\text{store}}/(3N^3)$
Spatial qubits, n